

funTDA — Part B: time series baseline (Tribolium beetle model)

Persistence landscapes, FPCA, and functional permutation tests over 400 simulated series

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Data

This is Part B of the funTDA project: the same persistence-landscape-based functional data analysis (FDA) pipeline used in Part A for gene regulatory networks, applied here to **simulated time series** instead. The system is the Tribolium flour beetle population model of Costantino et al. (1995). Pereira & de Mello (2015) is the reference for applying persistent homology to time series via Takens-delay-embedded point clouds, the general approach `beetles_landscapes.ipynb` follows here; this document has not independently verified that Pereira & de Mello’s own worked examples use the Tribolium model specifically, as opposed to some other time series.

The discrete-time model tracks three life stages (larvae L, pupae/callow adults P, mature adults A) in two-week intervals. Two parameter regimes are simulated, 200 series each, 201 time steps, with initial conditions $L_0, P_0, A_0 \sim U[2, 100]$:

- **Stable** ($u_a = 0.73$, adult death rate): the population converges to a fixed-point equilibrium.
- **Aperiodic** ($u_a = 0.96$): the population oscillates chaotically.

i This functional test is new, not a reproduction

Two things distinguish this document from a re-run of prior work:

1. **Higgins, Wu & Carey’s funTDA paper does not analyze time series at all.** Its worked examples are all networks — Erdős–Rényi graphs, stochastic block models, a literary character network, and the H3N2 gene regulatory networks used in Part A. Applying the funTDA landscape + FPCA + permutation-test pipeline to time series is an extension of the method to a data modality the original paper never covers.
2. **The original `Example1Beetles.R` does not run this functional hypothesis test either.** It runs two *independent* classification checks, each against the true regime labels, each reported as its own confusion matrix:
 - (a) k-means clustering of the raw (unnormalized) series into two groups, compared to the true labels (`cfm <- table(correct, km_clusters)`), and separately (b) a rule that calls a series “correctly” classified if its single most persistent bar has dimension 0 (stable) or 1 (aperiodic), again compared to the true labels (`cfm_hom`) — not to the k-means clusters from (a). The two checks are never compared to each other. There is no landscape, no FPCA, and no permutation test in it — both are classification-accuracy checks against ground truth, not a test of functional equality between group means.

The FPCA and permutation test below are therefore a genuinely new analysis, not a reproduction of a previously reported number — unlike Part A, there is no prior “baseline” p-value or variance-explained figure to match. The `ref` values in this document’s source are sanity checks from an earlier direct computation on these same committed arrays, not an external ground truth.

Pipeline: from simulated series to functional landscapes

The full pipeline — simulation, per-series min-max normalisation to $[0, 1]$, Takens delay embedding ($m = 2, \tau = 3$), Vietoris–Rips persistent homology via `ripser` (the same underlying algorithm `TDastats::calculate_homology` uses in R), and the first H1 persistence landscape layer ($KK = 1$) on a shared 500-point grid — is implemented in Python, in `beetles_landscapes.ipynb`. The grid there is **data-adaptive**: `tseq = linspace(global_min_birth, global_max_death, 500)`, rather than the fixed $[0, 1]$ Part A could use, because Rips filtration values on a normalised phase-space point cloud don’t have a natural upper bound of 1.

That notebook saves three arrays (`stable_landscapes.npy`, shape (200, 500); `aperiodic_landscapes.npy`, shape (200, 500); `tseq.npy`, shape (500,)). `export_landscapes.py` converts them to the CSVs this document reads (`output/stable_landscapes.csv`, `output/aperiodic_landscapes.csv`, `output/tseq.csv`) — mirroring Part A’s split between a Python stage that computes persistent homology and an R stage that does the functional data analysis. The landscapes are **not** re-simulated or re-computed here: doing so is slow (`ripser` over 400 point clouds), and Python’s and R’s random number generators aren’t interchangeable, so re-simulating in R would not reproduce these particular 400 series anyway. The committed CSVs are the single source of truth for everything that follows.

i Reused figures from `beetles_landscapes.ipynb`

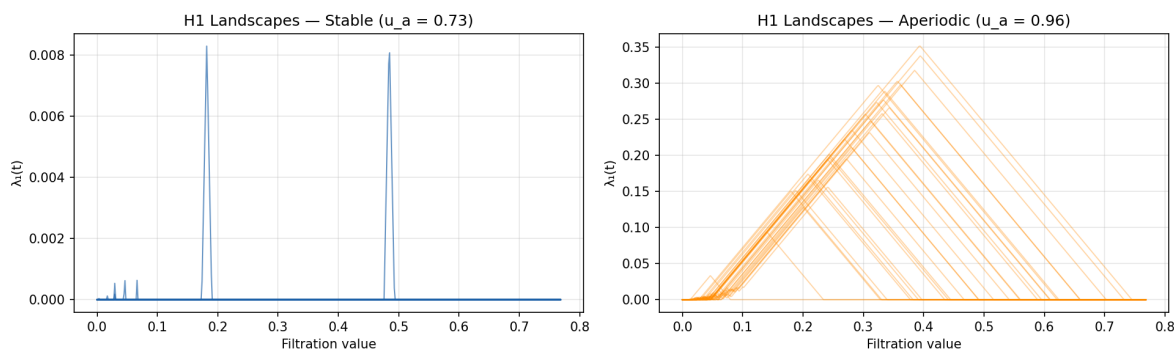


Figure 1: Individual H1 landscapes, 30 example series per regime (generated in the Python notebook, reused as-is).

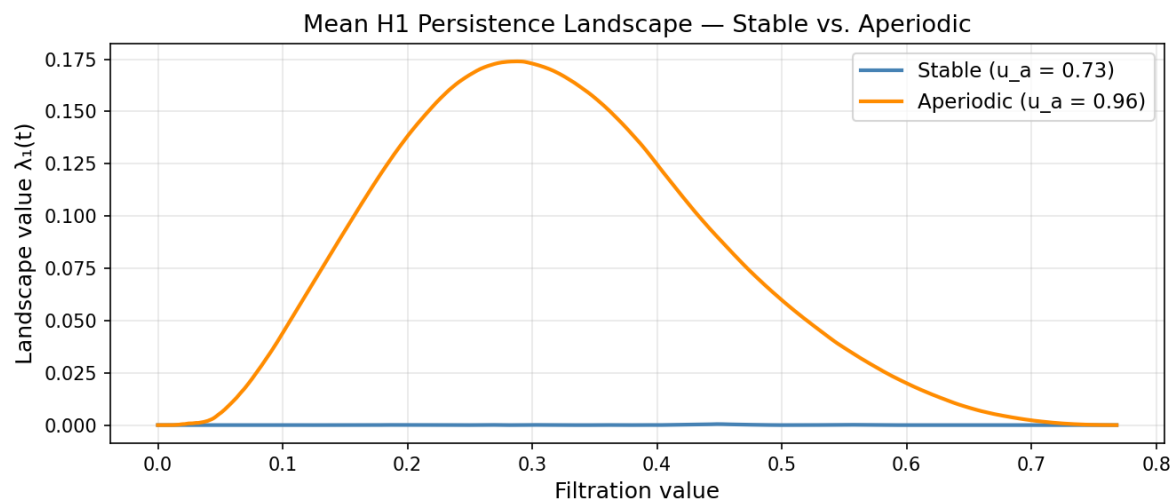


Figure 2: Mean H1 landscape per regime (generated in the Python notebook, reused as-is).

Structural check: most stable landscapes are identically zero

129 of the 200 stable series (64%) have a landscape that is identically zero across the whole grid, versus **0 of 200** aperiodic series. This is the expected topological signature of the two regimes, not noise: $u_a = 0.73$ drives the population to a fixed-point equilibrium, whose Takens-embedded attractor is a compact cluster with no loops — no H1 features, hence a landscape that is exactly zero everywhere. $u_a = 0.96$ drives sustained oscillation, whose attractor is a closed orbit — an H1 loop that the landscape detects directly. Consistent with this, the mean of each series' landscape peak is **0.0016** for stable series versus **0.2274** for aperiodic ones — roughly a 150-fold difference in typical peak height.

Functional Principal Component Analysis (FPCA)

Identical smoothing to `funTDA.R`: an order-2 (piecewise-linear) B-spline basis with `nbasis = 499` (one fewer than the 500 grid points), so the basis nearly interpolates the raw landscape values. The only difference from Part A is the basis range: `range(tseq)` here (the data-adaptive Rips scale, about $[0, 0.77]$) instead of the fixed $[0, 1]$ Part A could use. PC1’s sign is fixed by convention (stable series score negative) immediately after `pca.fd`, since an eigenvector’s sign is otherwise arbitrary — see the code below.

Table 1: Variance explained by the first two functional components

Component	% variance explained
PC1	84.29%
PC2	12.67%
Cumulative (PC1+PC2)	96.96%

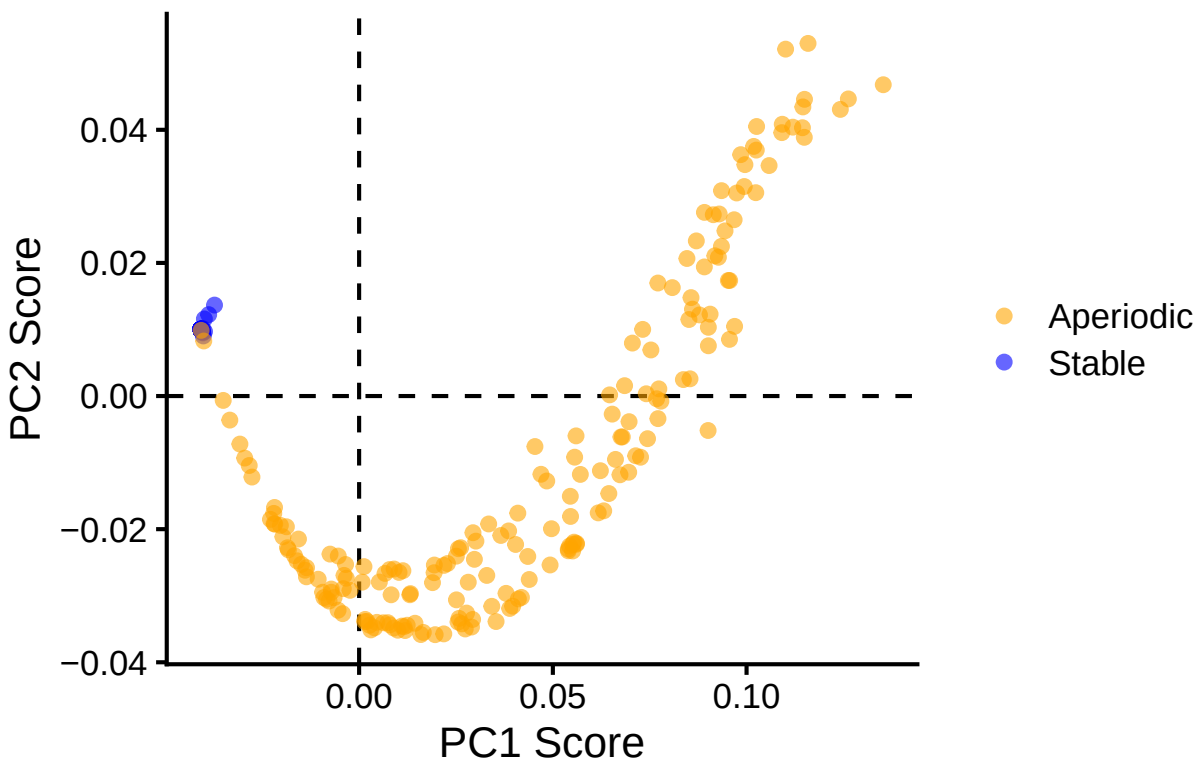


Figure 3: PC1 vs PC2 scores (FPCA over H1 landscapes), colored by regime. All 400 points are individual series; no per-point labels, unlike Part A’s 17-subject plot, since labeling 400 points would be unreadable — see the full score table below for per-series values.

i Full FPC scores (all 400 series)

ID	Group	PC1	PC2
Stable1	Stable	-0.0408	0.0100
Stable2	Stable	-0.0408	0.0100
Stable3	Stable	-0.0408	0.0100
Stable4	Stable	-0.0408	0.0100
Stable5	Stable	-0.0408	0.0100
Stable6	Stable	-0.0408	0.0100
Stable7	Stable	-0.0408	0.0100
Stable8	Stable	-0.0408	0.0100
Stable9	Stable	-0.0408	0.0100
Stable10	Stable	-0.0408	0.0100
Stable11	Stable	-0.0408	0.0100
Stable12	Stable	-0.0408	0.0100
Stable13	Stable	-0.0408	0.0100
Stable14	Stable	-0.0408	0.0100
Stable15	Stable	-0.0407	0.0101
Stable16	Stable	-0.0408	0.0100
Stable17	Stable	-0.0408	0.0100
Stable18	Stable	-0.0408	0.0100
Stable19	Stable	-0.0408	0.0100
Stable20	Stable	-0.0408	0.0100
Stable21	Stable	-0.0408	0.0100
Stable22	Stable	-0.0408	0.0100
Stable23	Stable	-0.0408	0.0100
Stable24	Stable	-0.0408	0.0100
Stable25	Stable	-0.0408	0.0100
Stable26	Stable	-0.0408	0.0100
Stable27	Stable	-0.0408	0.0100
Stable28	Stable	-0.0408	0.0100
Stable29	Stable	-0.0408	0.0100
Stable30	Stable	-0.0408	0.0100
Stable31	Stable	-0.0408	0.0100
Stable32	Stable	-0.0408	0.0100
Stable33	Stable	-0.0408	0.0101
Stable34	Stable	-0.0408	0.0100
Stable35	Stable	-0.0408	0.0100
Stable36	Stable	-0.0408	0.0100
Stable37	Stable	-0.0408	0.0100
Stable38	Stable	-0.0408	0.0100
Stable39	Stable	-0.0408	0.0100
Stable40	Stable	-0.0408	0.0100
Stable41	Stable	-0.0408	0.0100
Stable42	Stable	-0.0408	0.0100
Stable43	Stable	-0.0408	0.0100
Stable44	Stable	-0.0408	0.0100

Stable45	Stable	-0.0408	0.0100
Stable46	Stable	-0.0408	0.0100
Stable47	Stable	-0.0408	0.0100
Stable48	Stable	-0.0408	0.0100
Stable49	Stable	-0.0408	0.0100
Stable50	Stable	-0.0408	0.0100
Stable51	Stable	-0.0408	0.0100
Stable52	Stable	-0.0407	0.0101
Stable53	Stable	-0.0408	0.0100
Stable54	Stable	-0.0408	0.0100
Stable55	Stable	-0.0408	0.0100
Stable56	Stable	-0.0408	0.0100
Stable57	Stable	-0.0408	0.0100
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Stable76	Stable	-0.0404	0.0101
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Stable79	Stable	-0.0408	0.0100
Stable80	Stable	-0.0408	0.0100
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Aperiodic113	Aperiodic	0.1144	0.0403
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Aperiodic120	Aperiodic	0.0956	0.0085

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Aperiodic134	Aperiodic	0.0555	-0.0220
Aperiodic135	Aperiodic	0.0219	-0.0254
Aperiodic136	Aperiodic	0.0753	0.0069
Aperiodic137	Aperiodic	0.0396	-0.0316
Aperiodic138	Aperiodic	-0.0078	-0.0308
Aperiodic139	Aperiodic	0.0411	-0.0305
Aperiodic140	Aperiodic	0.0131	-0.0299
Aperiodic141	Aperiodic	0.0410	-0.0176
Aperiodic142	Aperiodic	0.1092	0.0408
Aperiodic143	Aperiodic	0.0110	-0.0346
Aperiodic144	Aperiodic	0.0074	-0.0340
Aperiodic145	Aperiodic	0.0079	-0.0344
Aperiodic146	Aperiodic	-0.0277	-0.0122
Aperiodic147	Aperiodic	0.0386	-0.0203
Aperiodic148	Aperiodic	0.0545	-0.0151
Aperiodic149	Aperiodic	0.0379	-0.0296
Aperiodic150	Aperiodic	0.1026	0.0405
Aperiodic151	Aperiodic	-0.0084	-0.0305
Aperiodic152	Aperiodic	-0.0072	-0.0290
Aperiodic153	Aperiodic	-0.0401	0.0083
Aperiodic154	Aperiodic	0.0256	-0.0229
Aperiodic155	Aperiodic	0.0258	-0.0334
Aperiodic156	Aperiodic	0.0779	-0.0008
Aperiodic157	Aperiodic	0.0132	-0.0297
Aperiodic158	Aperiodic	0.0262	-0.0228
Aperiodic159	Aperiodic	-0.0091	-0.0303
Aperiodic160	Aperiodic	0.0052	-0.0279
Aperiodic161	Aperiodic	0.0631	-0.0172
Aperiodic162	Aperiodic	0.1101	0.0521
Aperiodic163	Aperiodic	0.0251	-0.0306
Aperiodic164	Aperiodic	0.0714	-0.0090
Aperiodic165	Aperiodic	0.0496	-0.0199
Aperiodic166	Aperiodic	0.0100	-0.0352

Aperiodic167	Aperiodic	0.0116	-0.0347
Aperiodic168	Aperiodic	0.0228	-0.0252
Aperiodic169	Aperiodic	0.0255	-0.0340
Aperiodic170	Aperiodic	0.0935	0.0309
Aperiodic171	Aperiodic	-0.0308	-0.0072
Aperiodic172	Aperiodic	0.0891	0.0276
Aperiodic173	Aperiodic	0.0676	-0.0062
Aperiodic174	Aperiodic	-0.0106	-0.0275
Aperiodic175	Aperiodic	0.0551	-0.0233
Aperiodic176	Aperiodic	0.0294	-0.0205
Aperiodic177	Aperiodic	0.0389	-0.0319
Aperiodic178	Aperiodic	0.0103	-0.0265
Aperiodic179	Aperiodic	0.0968	0.0265
Aperiodic180	Aperiodic	0.0653	-0.0027
Aperiodic181	Aperiodic	0.0282	-0.0280
Aperiodic182	Aperiodic	0.0645	-0.0146
Aperiodic183	Aperiodic	0.0697	-0.0039
Aperiodic184	Aperiodic	0.0726	-0.0092
Aperiodic185	Aperiodic	0.0484	-0.0128
Aperiodic186	Aperiodic	-0.0035	-0.0253
Aperiodic187	Aperiodic	0.0679	-0.0061
Aperiodic188	Aperiodic	-0.0218	-0.0193
Aperiodic189	Aperiodic	0.0846	0.0207
Aperiodic190	Aperiodic	-0.0188	-0.0196
Aperiodic191	Aperiodic	0.0046	-0.0340
Aperiodic192	Aperiodic	0.0685	0.0016
Aperiodic193	Aperiodic	0.0265	-0.0342
Aperiodic194	Aperiodic	-0.0220	-0.0176
Aperiodic195	Aperiodic	-0.0023	-0.0292
Aperiodic196	Aperiodic	0.0251	-0.0241
Aperiodic197	Aperiodic	0.0551	-0.0223
Aperiodic198	Aperiodic	-0.0149	-0.0253
Aperiodic199	Aperiodic	0.0906	0.0123
Aperiodic200	Aperiodic	0.1159	0.0530

i Sanity checks — FPCA

Metric	Sanity-check value	Computed here
PC1	~84%	84.29%
PC2	~13%	12.67%
Cumulative	~97%	96.96%
Stable with PC1 < 0	200 / 200	200 / 200
Aperiodic with PC1 < 0	~42 / 200	42 / 200

The first two functional components explain **97.0%** of the variability among landscapes (PC1 $\approx$

84.3%, $PC2 \approx 12.7\%$). All 200 stable series fall in the $PC1 < 0$ half-plane — unsurprising, since 129 of them share the exact same (identically-zero) landscape and so collapse to nearly the same point in function space. Separation is **not** perfect, though: **42 of the 200 aperiodic series** (21%) also fall in $PC1 < 0$, overlapping the stable cloud.

This is a genuine limitation of the fixed embedding parameters ($m = 2, \tau = 3$), not of the FDA pipeline: for some aperiodic trajectories, this particular embedding does not produce a topologically prominent loop — the reconstructed attractor stays close enough to a compact blob that its H1 landscape resembles a stable series’ near-zero landscape. Takens’ theorem guarantees a *generic* embedding recovers the attractor topology, but (m, τ) are fixed constants here, not tuned per series; a natural extension would be a sensitivity analysis over (m, τ) to check whether a higher embedding dimension or a different delay recovers the loop for these borderline series.

Functional permutation test: stable vs. aperiodic

This reuses the same vectorized max- T permutation test from Part A — already validated bit-for-bit there against `fda::tperm.fd` — evaluating each `fd` object on a 101-point grid and permuting columns of a matrix instead of re-evaluating `fd` on every permutation. The masked version below is mathematically identical to that validated function whenever every grid point’s denominator is positive: that is the case for every grid point in every test Part A ever ran (its GRN landscapes never included an identically-zero one), so Part A’s bit-for-bit validation carries over directly to those points. What is new here is a defensive guard for a case Part A’s data never exercised.

! A defensive guard for a zero-variance denominator (does not trigger with this data)

The Welch-style test statistic at each grid point t is $|\bar{x}_1(t) - \bar{x}_2(t)| / \sqrt{v_1(t) + v_2(t)}$, and the final statistic is the supremum over t . Because 129 of the 200 stable series are identically zero, the stable group’s pointwise variance $v_1(t)$ is minuscule almost everywhere, and at the very edges of the shared grid range — where essentially no series in *either* group has accumulated any persistence yet — both groups’ variances could in principle hit exact floating-point zero, making the ratio the undefined form 0/0. Part A’s networks never produced an identically-zero landscape, so this case never had to be guarded against there.

The guard excludes grid points with a zero denominator from the supremum, rather than let a stray NaN or Inf propagate into `max()` unnoticed:

```
den <- v1 + v2
valid <- den > 0
max(abs(m1[valid] - m2[valid]) / sqrt(den[valid]))
```

With the actual committed data, this guard turns out **not to be needed** in practice (confirmed just below, computed live from the same objects the test uses): after B-spline smoothing, the boundary points evaluate to tiny floating-point residues (of order 10^{-13} to 10^{-88}) rather than exact zero, so they pass the `den > 0` filter and contribute a small, well-defined statistic value rather than an undefined one. The guard is kept anyway, as a correctness safeguard rather than a currently-active code path: a slightly different smoothing, grid, or platform could in principle produce an exact zero, and silently propagating a NaN into a `max()` would be a much worse failure mode than an unused `if`.

! Stable vs. Aperiodic test result

- **Observed $T_{\max}$** = 85.51 (sanity check: ~ 85.5)
- **Critical value (95%)** = 2.75 (sanity check: ~ 2.85)
- **p-value** = 0.00000 (out of 10000 permutations — a value of 0 here means $p < 1e-4$, not exactly zero)

With $T_{\max} = 85.51$ — dramatically above the 95% critical value of 2.75 — the null hypothesis of functional equality between the mean stable and aperiodic H1 landscapes is rejected outright. Note that $T_{\max}$ is **not** an effect size: the supremum is reached at $t \approx 0.1844$ (mean difference ≈ 0.125), not at $t \approx 0.2842$ where the mean difference between regimes is actually largest (≈ 0.174 , giving a statistic of only 35.95) — because the statistic is studentized, dividing by the local pooled variance, so it is largest where the groups are both tightly concentrated *and* different, not simply where they differ most in absolute terms. The evidence for a real difference is the p-value and the mean-landscape comparison (Figure 2 above), not the raw scale of $T_{\max}$. As noted above, the zero-variance guard was not actually needed here: 101 of 101 grid points are valid for this test (computed live from the same objects `vtperm` uses), and the within-regime control splits below are likewise always fully valid.

This is expected given how the two regimes were constructed (one produces topological loops, the other does not), but the result itself is new: `Example1Beetles.R` never asked this question in a functional-hypothesis-testing sense. It runs two independent classification checks against the true regime labels — k-means clustering on the raw series (`cfm`) and a rule based on which homology dimension each series' single most persistent bar belongs to (`cfm_hom`) — and reports each as its own confusion matrix; the two checks are never compared against each other, and neither is a test of functional equality between group means. The permutation test here instead directly tests whether the *entire mean landscape curves* differ, which is the same kind of claim Part A makes about asymptomatic vs. symptomatic GRNs.

Control analysis (within-regime calibration)

Part A's split-half validation enumerated *every* possible partition of each group (`choose(9,4) = 126`, `choose(8,4) = 70`) because those groups were small enough to do so exhaustively. Here each regime has 200 series, and `choose(200, 100)` is on the order of 10^{58} — completely infeasible to enumerate. Instead, this is a **random sample of 200 balanced (100 vs. 100) splits per regime**, drawn with a fixed seed continuing directly from the main test's RNG stream above (not “all partitions”, unlike Part A). Because these splits are all drawn from the same 200 series per regime (not 200 independent fresh samples), they are correlated with each other, so the mean/sd/fraction below understate the true sampling uncertainty somewhat — the same caveat applies, in a milder form, to Part A's exhaustive enumeration, since overlapping subsets there are not independent either.

i The stable control is not as informative as the aperiodic one

The two regimes' control analyses are not equivalent evidence. 129 of the 200 stable series (64%) are identically zero, so a random 100-vs-100 split of the stable regime compares two halves that are, on average, **64%** identical zero curves each (computed live across the 200 splits

actually drawn above) — it is substantially a “zeros vs. zeros” comparison, well calibrated close to trivially rather than as a demonstration that the test correctly handles genuine functional variability. The zero-variance guard from the previous section is, in fact, never triggered in these splits either: all 101 of 101 grid points are always valid (computed live, same as the main test above). The **aperiodic control is the informative one**: no aperiodic series is identically zero, so its 200 splits compare genuinely varying curves on the full, unmasked grid (101 of 101 points valid there too, for the unrelated reason that no aperiodic series has zero variance in the first place) — its calibration result is the one that should be read as evidence the test does not over-reject.

Table 2: Summary of the within-regime control analysis

Regime	mean		
	p-value	sd(p-value)	% splits with $p < 0.05$
Stable (200 random splits, 100 vs 100)	0.534	0.297	5.0%
Aperiodic (200 random splits, 100 vs 100)	0.505	0.282	4.0%

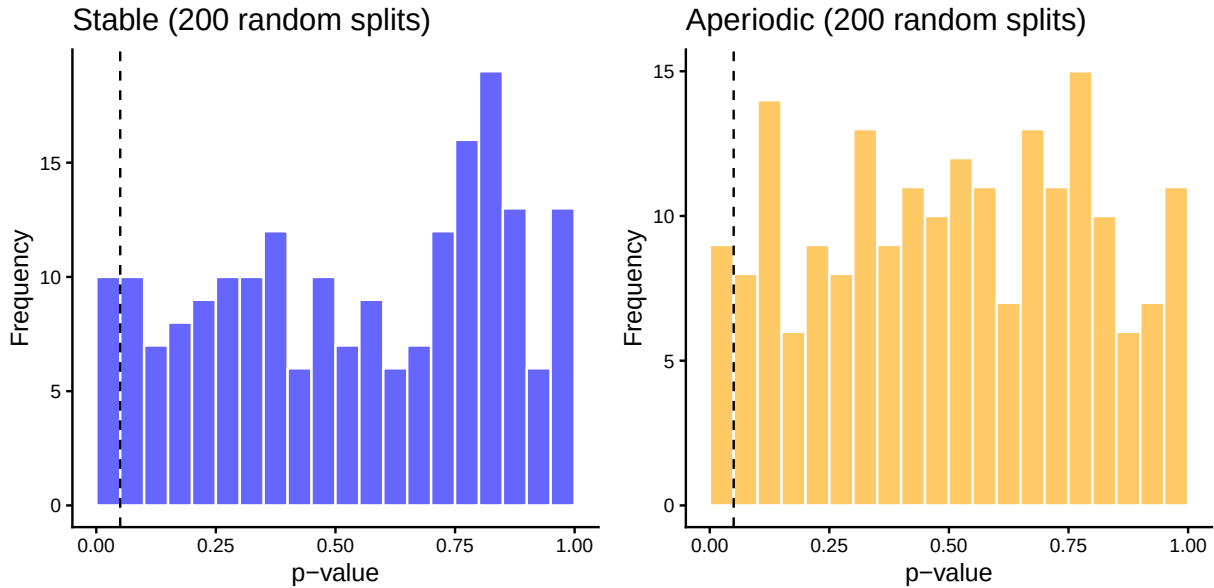


Figure 4: Distribution of control-analysis p-values, by regime (dashed line at $p = 0.05$).

i Sanity checks — control analysis

Regime	mean p	sd p	% $p < 0.05$
Stable	0.534	0.297	5.0%
Aperiodic	0.505	0.282	4.0%

Both regimes' p-values are consistent with the nominal 5% false-positive rate expected under the null, but as noted above, only the **aperiodic** result is strong evidence that the test does not over-reject on genuinely varying functional data — the stable result is largely a check that comparing zero to zero correctly fails to reject, which is a much lower bar. Taken together, they still calibrate the significance of the stable-vs-aperiodic result above: the rejection there reflects a real difference between regimes, not a general tendency of the test to over-reject.

Appendix

Session information

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
```

```
Platform: aarch64-apple-darwin20
```

```
Running under: macOS Tahoe 26.5.2
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK
```

```
locale:
```

```
[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
```

```
time zone: Europe/Madrid
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] splines stats graphics grDevices utils datasets methods
```

```
[8] base
```

```
other attached packages:
```

```
[1] patchwork_1.3.2 ggplot2_4.0.3 fda_6.3.0 deSolve_1.42
```

```
[5] fds_1.9 RCurl_1.98-1.19 rainbow_3.8 pcaPP_2.0-5
```

```
[9] MASS_7.3-65
```

```
loaded via a namespace (and not attached):
```

```
[1] Matrix_1.7-4 gtable_0.3.6 jsonlite_2.0.0 dplyr_1.2.1
[5] compiler_4.5.3 tidyselect_1.2.1 ks_1.15.2 bitops_1.0-9
[9] cluster_2.1.8.2 scales_1.4.0 yaml_2.3.12 fastmap_1.2.0
[13] lattice_0.22-9 R6_2.6.1 labeling_0.4.3 generics_0.1.4
[17] knitr_1.51 tibble_3.3.1 pillar_1.11.1 RColorBrewer_1.1-3
[21] rlang_1.2.0 hdrclde_3.5.0 xfun_0.57 S7_0.2.2
[25] cli_3.6.6 withr_3.0.2 magrittr_2.0.5 digest_0.6.39
[29] grid_4.5.3 mvtnorm_1.4-2 mclust_6.1.3 lifecycle_1.0.5
```

[33] vctrs_0.7.3	KernSmooth_2.23-26	evaluate_1.0.5	pracma_2.4.6
[37] glue_1.8.1	farver_2.1.2	colorspace_2.1-3	rmarkdown_2.31
[41] pkgconfig_2.0.3	tools_4.5.3	htmltools_0.5.9	

Full code (reproducible)

```
library(fda)
library(ggplot2)
library(patchwork)

# Reference values from a prior direct computation on the committed landscape
# matrices (for sanity-check comparison only; every number reported in the
# body is recomputed here from output/*.csv, not hardcoded from this list).
ref <- list(
  n_stable_zero = 129, n_aperiodic_zero = 0,
  mean_max_stable = 0.0016, mean_max_aperiodic = 0.227,
  pc1_pct = 84, pc2_pct = 13,
  n_aperiodic_pc1_neg = 42,
  test1_Tobs = 85.5, test1_crit95 = 2.85
)

col_stable <- "blue"
col_aperiodic <- "orange"
tseq <- as.matrix(read.csv("output/tseq.csv", header = FALSE))[, 1]
DataStable <- as.matrix(read.csv("output/stable_landscapes.csv", header = FALSE))
DataAperiodic <- as.matrix(read.csv("output/aperiodic_landscapes.csv", header = FALSE))
DataBoth <- rbind(DataStable, DataAperiodic)

stopifnot(
  "Expected 200 stable series" == nrow(DataStable) == 200,
  "Expected 200 aperiodic series" == nrow(DataAperiodic) == 200,
  "Expected a 500-point shared grid" == length(tseq) == 500,
  "Landscape matrices must have one column per grid point" =
    ncol(DataStable) == length(tseq) && ncol(DataAperiodic) == length(tseq)
)

is_zero_row <- function(M) apply(M, 1, function(r) all(r == 0))
n_stable_zero <- sum(is_zero_row(DataStable))
n_aperiodic_zero <- sum(is_zero_row(DataAperiodic))

stopifnot(
  "Expected exactly 129 stable series with an identically-zero landscape" =
    n_stable_zero == 129,
  "Expected 0 aperiodic series with an identically-zero landscape" =
    n_aperiodic_zero == 0
)
```

```

mean_max_stable      <- mean(apply(DataStable, 1, max))
mean_max_aperiodic  <- mean(apply(DataAperiodic, 1, max))
tol_meanmax <- 0.01
stopifnot(
  "Mean stable landscape peak out of tolerance"      = abs(mean_max_stable - ref$mean_max_stable) < tol_meanmax,
  "Mean aperiodic landscape peak out of tolerance" = abs(mean_max_aperiodic - ref$mean_max_aperiodic) < tol_meanmax,
)
create_fd <- function(Data, tseq) {
  basis <- create.bspline.basis(range(tseq), nbasis = ncol(Data) - 1, norder = 2)
  fdobj <- smooth.basis(tseq, t(Data), basis)
  fdobj$fd
}

fdStable      <- create_fd(DataStable, tseq)
fdAperiodic <- create_fd(DataAperiodic, tseq)
fdBoth       <- create_fd(DataBoth, tseq)

pca <- pca.fd(fdBoth, nharm = 2)
pc1_pct <- pca$varprop[1] * 100
pc2_pct <- pca$varprop[2] * 100
cum_pct <- pc1_pct + pc2_pct

# An eigenvector's sign is mathematically arbitrary (pca.fd can return either
# sign depending on the LAPACK/fda version), so PC1 alone is not a stable
# quantity to assert on across environments. Fix it deterministically here,
# before computing any score or assert, to the convention "stable series
# score negative on PC1" - an independent PCA run on these same arrays has
# been observed to return the opposite sign, so this is not hypothetical.
pc1_raw <- pca$scores[, 1]
sign_fix <- if (mean(pc1_raw[1:200]) > 0) -1 else 1 # 1:200 = stable rows
PC1 <- sign_fix * pc1_raw
PC2 <- pca$scores[, 2] # PC2's sign is independently arbitrary but no assert depends on it

ids <- c(paste0("Stable", 1:200), paste0("Aperiodic", 1:200))
group <- c(rep("Stable", 200), rep("Aperiodic", 200))

pca_df <- data.frame(
  PC1 = PC1,
  PC2 = PC2,
  ID = ids,
  Group = group
)

n_stable_pc1_neg <- sum(pca_df$PC1[pca_df$Group == "Stable"] < 0)
n_aperiodic_pc1_neg <- sum(pca_df$PC1[pca_df$Group == "Aperiodic"] < 0)
knitr::kable(
  data.frame(

```

```

    Component = c("PC1", "PC2", "Cumulative (PC1+PC2)"),
    `"% variance explained"` = c(sprintf("%.2f%%", pc1_pct), sprintf("%.2f%%", pc2_pct), sprintf(
    check.names = FALSE
  ),
  align = "lr"
)
ggplot(pca_df, aes(x = PC1, y = PC2, color = Group)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "black") +
  geom_point(size = 2, alpha = 0.6) +
  scale_color_manual(values = c("Stable" = col_stable, "Aperiodic" = col_aperiodic)) +
  labs(x = "PC1 Score", y = "PC2 Score", title = "") +
  theme_classic(base_size = 16) +
  theme(
    legend.position = "right",
    legend.title = element_blank(),
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
pca_full_df <- data.frame(
  ID = pca_df$ID,
  Group = pca_df$Group,
  PC1 = round(pca_df$PC1, 4),
  PC2 = round(pca_df$PC2, 4)
)
knitr::kable(pca_full_df, align = "llrr")
tol_fpca <- 2 # percentage points - this is a new analysis, not a reproduction, hence a loose
stopifnot(
  "PC1 far from sanity-check value" = abs(pc1_pct - ref$pc1_pct) < tol_fpca,
  "PC2 far from sanity-check value" = abs(pc2_pct - ref$pc2_pct) < tol_fpca,
  "Expected all 200 stable series to have PC1 < 0" = n_stable_pc1_neg == 200,
  "Aperiodic PC1<0 count far from sanity-check value" = abs(n_aperiodic_pc1_neg - ref$n_aperiodic_pc1_neg) < tol_fpca
)
rowVar <- function(M) {
  n <- ncol(M)
  mu <- rowMeans(M)
  rowSums((M - mu)^2) / (n - 1)
}

vtperm <- function(X, i1, i2, nperm = 10000) {
  x1 <- X[, i1, drop = FALSE]
  x2 <- X[, i2, drop = FALSE]
  n1 <- ncol(x1); n2 <- ncol(x2)
  Xmat <- cbind(x1, x2)
  n <- n1 + n2
  Ts <- function(M) {
    m1 <- rowMeans(M[, 1:n1, drop = FALSE])
    m2 <- rowMeans(M[, n1 + (1:n2), drop = FALSE])
  }
}

```

```

    v1 <- rowVar(M[, 1:n1, drop = FALSE]) / n1
    v2 <- rowVar(M[, n1 + (1:n2), drop = FALSE]) / n2
    den <- v1 + v2
    valid <- den > 0
    max(abs(m1[valid] - m2[valid]) / sqrt(den[valid]))
  }
  Tobs <- Ts(Xmat)
  Tn <- numeric(nperm)
  for (i in 1:nperm) Tn[i] <- Ts(Xmat[, sample(n)])
  list(pval = mean(Tobs < Tn), Tobs = Tobs, crit95 = unname(quantile(Tn, 0.95)))
}

argvals <- seq(min(tseq), max(tseq), length.out = 101)
# The cache is keyed on a hash of its own inputs: if the committed CSVs ever
# change, the stored p-values would otherwise silently go stale and the
# asserts below would pass against outdated numbers instead of catching the
# change. tools::md5sum is base R, no extra dependency.
input_files <- c("output/tseq.csv", "output/stable_landscapes.csv", "output/aperiodic_landscapes.csv")
input_hash <- paste(unname(tools::md5sum(input_files)), collapse = "_")

cache_file <- "output/results_partB_permtest_cache.rds"
cached <- if (file.exists(cache_file)) readRDS(cache_file) else NULL
cache_valid <- !is.null(cached) && identical(cached$input_hash, input_hash)

count_valid <- function(v1, v2) sum((v1 + v2) > 0)

if (!cache_valid) {
  set.seed(123)

  # 1) Main test: Stable vs Aperiodic
  XFull <- cbind(eval.fd(argvals, fdStable), eval.fd(argvals, fdAperiodic))
  test1 <- vtperm(XFull, 1:200, 201:400, nperm = 10000)

  # Diagnostics for the main test (descriptive only; does not affect test1
  # or consume any random numbers, so it cannot change which permutations
  # get drawn below).
  m1_diag <- rowMeans(XFull[, 1:200]); m2_diag <- rowMeans(XFull[, 201:400])
  v1_diag <- rowVar(XFull[, 1:200]) / 200; v2_diag <- rowVar(XFull[, 201:400]) / 200
  den_diag <- v1_diag + v2_diag
  diff_diag <- abs(m1_diag - m2_diag)
  stat_diag <- ifelse(den_diag > 0, diff_diag / sqrt(den_diag), NA_real_)
  i_sup <- which.max(stat_diag)
  i_diffmax <- which.max(diff_diag)
  test1_diag <- list(
    n_valid = count_valid(v1_diag, v2_diag), n_grid = length(den_diag),
    t_sup = argvals[i_sup], diff_sup = diff_diag[i_sup], stat_sup = stat_diag[i_sup],
    t_diffmax = argvals[i_diffmax], diff_diffmax = diff_diag[i_diffmax], stat_diffmax = stat_d

```

```

)

# 2) Control analysis, stable regime: exhaustive enumeration of all 4-vs-4-
# style splits (as in Part A) is infeasible here (choose(200,100) is
# astronomically large), so instead draw a random sample of 200 balanced
# (100 vs 100) splits. Alongside each split's p-value, also record (a)
# how many of the 101 grid points survive the zero-variance mask and
# (b) what fraction of each half's 100 series is identically zero -
# neither consumes random numbers, so neither changes pvalsStable.
XStable <- eval.fd(argvals, fdStable)
n_splits <- 200
pvalsStable <- numeric(n_splits)
validStable <- integer(n_splits)
fracZeroStable <- numeric(n_splits)
for (i in seq_len(n_splits)) {
  g1 <- sample(1:200, 100); g2 <- setdiff(1:200, g1)
  pvalsStable[i] <- vtperm(XStable, g1, g2, nperm = 10000)$pval
  vv1 <- rowVar(XStable[, g1]) / 100; vv2 <- rowVar(XStable[, g2]) / 100
  validStable[i] <- count_valid(vv1, vv2)
  fracZeroStable[i] <- mean(c(sum(is_zero_row(DataStable[g1, , drop = FALSE])),
                             sum(is_zero_row(DataStable[g2, , drop = FALSE])))) / 100
}

# 3) Control analysis, aperiodic regime: same random-sample approach and
# the same grid-validity diagnostic (no aperiodic series is identically
# zero, so there is no zero-fraction to report here).
XAperiodic <- eval.fd(argvals, fdAperiodic)
pvalsAperiodic <- numeric(n_splits)
validAperiodic <- integer(n_splits)
for (i in seq_len(n_splits)) {
  g1 <- sample(1:200, 100); g2 <- setdiff(1:200, g1)
  pvalsAperiodic[i] <- vtperm(XAperiodic, g1, g2, nperm = 10000)$pval
  vv1 <- rowVar(XAperiodic[, g1]) / 100; vv2 <- rowVar(XAperiodic[, g2]) / 100
  validAperiodic[i] <- count_valid(vv1, vv2)
}

cached <- list(
  input_hash = input_hash, test1 = test1, test1_diag = test1_diag,
  pvalsStable = pvalsStable, pvalsAperiodic = pvalsAperiodic,
  validStable = validStable, validAperiodic = validAperiodic,
  fracZeroStable = fracZeroStable
)
saveRDS(cached, cache_file)
}

test1 <- cached$test1
test1_diag <- cached$test1_diag

```

```

pvalsStable      <- cached$pvalsStable
pvalsAperiodic   <- cached$pvalsAperiodic
validStable      <- cached$validStable
validAperiodic   <- cached$validAperiodic
fracZeroStable   <- cached$fracZeroStable

stopifnot(
  "Expected 200 stable control splits"      = length(pvalsStable) == 200,
  "Expected 200 aperiodic control splits" = length(pvalsAperiodic) == 200
)
tol_Tobs      <- 15 # loose: Tobs is a studentized statistic, sensitive to which grid point the
tol_crit95    <- 0.5 # tighter: this is a quantile of 10000 permutation draws, not a studentized
stopifnot(
  "Observed Tmax far from sanity-check value" = abs(test1$Tobs - ref$test1_Tobs) < tol_Tobs,
  "95% critical value far from sanity-check value" = abs(test1$crit95 - ref$test1_crit95) < tol_Tobs,
  "Expected a highly significant result (p < 0.01)" = test1$pval < 0.01
)
splithalf_summary <- data.frame(
  Regime = c("Stable (200 random splits, 100 vs 100)", "Aperiodic (200 random splits, 100 vs 100)"),
  `mean p-value` = sprintf("%.3f", c(mean(pvalsStable), mean(pvalsAperiodic))),
  `sd(p-value)` = sprintf("%.3f", c(sd(pvalsStable), sd(pvalsAperiodic))),
  `% splits with p < 0.05` = sprintf("%.1f%", c(mean(pvalsStable < 0.05), mean(pvalsAperiodic < 0.05))),
  check.names = FALSE
)
knitr::kable(splithalf_summary, align = "lrrr")
histStable <- ggplot(data.frame(p = pvalsStable), aes(x = p)) +
  geom_histogram(binwidth = 0.05, boundary = 0, fill = col_stable, alpha = 0.6, color = "white") +
  geom_vline(xintercept = 0.05, linetype = "dashed", color = "black") +
  labs(title = "Stable (200 random splits)", x = "p-value", y = "Frequency") +
  theme_classic(base_size = 13)

histAperiodic <- ggplot(data.frame(p = pvalsAperiodic), aes(x = p)) +
  geom_histogram(binwidth = 0.05, boundary = 0, fill = col_aperiodic, alpha = 0.6, color = "white") +
  geom_vline(xintercept = 0.05, linetype = "dashed", color = "black") +
  labs(title = "Aperiodic (200 random splits)", x = "p-value", y = "Frequency") +
  theme_classic(base_size = 13)

histStable + histAperiodic
tol_mean <- 0.05
tol_frac <- 5 # percentage points - 200 random splits is a noisier estimate than Part A's exhaustive search
stopifnot(
  "Stable control mean p implausible"      = abs(mean(pvalsStable) - 0.5) < tol_mean + 0.1,
  "Aperiodic control mean p implausible" = abs(mean(pvalsAperiodic) - 0.5) < tol_mean + 0.1,
  "Stable control frac<0.05 far from nominal 5%" = abs(mean(pvalsStable < 0.05) * 100 - 5) < tol_frac,
  "Aperiodic control frac<0.05 far from nominal 5%" = abs(mean(pvalsAperiodic < 0.05) * 100 - 5) < tol_frac
)
sessionInfo()

```